Abigale Kim

Curriculum Vitæ

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Department of Computer Sciences
University of Wisconsin–Madison

Education

2024–on **PhD in Computer Science**, University of Wisconsin–Madison, Advisor: Prof. Xiangyao Yu

Relevant Coursework: Foundations of Data Management, High Performance Computing for Applications in Engineering, Game Theory, Optimization & Learning, Foundation Models

2023–2024 **MSc in Computer Science**, Carnegie Mellon University, Advisor: Prof. Andrew Pavlo

Relevant Coursework: Advanced Database Systems, Advanced Operating Systems and Distributed Systems, Deep Learning Systems, Computer Architecture

2018–2021 **BSc in Computer Science**, Carnegie Mellon University, Graduated with University Honors

Relevant Coursework: Operating System Design and Implementation, Database Systems, Optimizing Compilers, Introduction to Computer Security, Computer Graphics

Research Experience

2024–on **University of Wisconsin–Madison**, PhD researcher, Supervisor: Prof. Xiangyao Yu

Researching GPU-accelerated database systems, string matching, and memory compaction.

2023–2024 **Carnegie Mellon University**, Master's researcher, Supervisor: Prof. Andrew Pavlo

Researching the database system extensibility ecosystem.

2020–2021 **Carnegie Mellon University**, Undergraduate researcher, Supervisor: Prof. Andrew Pavlo

Utilizing just-in-time (JIT) compilation techniques to increase index key comparator performance.

2019–2020 **Carnegie Mellon University**, Undergraduate researcher, Supervisor: Prof. Afsaneh Doryab

Researched the correlation between student performance and lifestyle factors, using data collected from smart devices.

Publications

2026 HetMatch: GPU-Accelerated Heterogeneous String Pattern Matching. **Abigale Kim**, Xiangyao Yu. ADMS @ VLDB 2026.

2026 Rethinking Analytical Processing in the GPU Era. Bobbi Yogatama, Yifei Yang, Kevin Kristensen, Devesh Sarda, **Abigale Kim**, Adrian Cockcroft, Yu Teng, Joshua Patterson, Gregory Kimball, Wes McKinney, Weiwei Gong, Xiangyao Yu. CIDR 2026.

2025 Anarchy in the Database: A Survey and Evaluation of Database Management System Extensibility. **Abigale Kim**, Marco Slot, David G. Andersen, Andrew Pavlo. VLDB 2025.

2021 Understanding Health and Behavioral Trends of Successful Students Through Machine Learning Models. **Abigale Kim**, Fateme Nikseresht, Janine Dutcher, Michael Tumminia, Daniella Villalba, Sheldon Cohen, Kasey Creswel, J. Creswell, Anind Dey, Jennifer Mankoff, and Afsaneh Doryab. IHIET-AI 2021.

Talks

2025 What PostgreSQL Extensibility Can Teach Us About Composable Data Management Systems, CDMS @ VLDB 2025. [Slides]

2025 Anarchy in the Database: A Survey and Evaluation of Database Management System Extensibility, DuckDB in Science Meetup 2025. [Slides]

2025 Anarchy in the Database: A Survey and Evaluation of Database Management System Extensibility, VLDB 2025. [Slides]

2024 Anarchy in the Database: A Survey and Evaluation of Database Management System Extensibility, PGConf.dev 2024. [Slides] [Video]

2023 Database Management System Extensibility, Parallel Data Retreat 2024. [Slides]

Posters

2026 GPU Accelerated Heterogeneous String Pattern Matching. Midwest Data Day 2026. [Poster]

2025 Anarchy in the Database: A Survey and Evaluation of Database Management System Extensibility, VLDB 2025. [Poster]

2025 GPU Databases - The New Modality of Database Analytics, University of Wisconsin–Madison, Prospective Student Visit Days 2025. [Poster]

2023 Survey of Database Management System Extensibility, Parallel Data Retreat, 2023. [Poster]

Experience

2026-on **NVIDIA**, Systems Software Engineering Intern
Worked on GPU-accelerated string and Parquet Variant processing with the cuDF team.

2024-2024 **TileDB Inc.**, Software Engineer
Developed a Rust API for the TileDB storage engine.

2022-2023 **TileDB Inc.**, Software Engineer
Redesigned query engine to support complex predicates. Worked on data pipelining features.

2021-2021 **YugabyteDB**, Software Engineering Intern
Implemented cluster mapping-update integration during asynchronous tablet split.

2020-2020 **Amazon Web Services, Redshift**, Software Development Engineering Intern
Designed and implemented a scalable service improving Redshift database cold-start performance.

2019-2019 **Amazon**, Future Engineer Software Development Engineering Intern
Developed web publishing platform used by 6,000 Amazon content merchandisers.

Awards

2025 **First Year Summer Fellowship**, Awarded summer funding to academically exceptional first year PhD students.

2022 **Senior Leadership Recognition**, Awarded to graduating seniors from Carnegie Mellon University who have made an unparalleled impact on the CMU community.

2022 **University Honors**, Awarded to graduating seniors from Carnegie Mellon University who graduated with a GPA above 3.5.

2021 **Michael Blaine Burks Scholarship**, Merit scholarship awarded to one student per year for demonstrated intellectual curiosity and perseverance in computer science.

Teaching

2026-2026 **University of Wisconsin–Madison**, Database Systems Teaching Assistant

2023-2023 **Carnegie Mellon University**, Database Systems Teaching Assistant

2021-2021 **Carnegie Mellon University**, Database Systems Teaching Assistant

2021-2021 **Carnegie Mellon University**, Introduction to Computer Systems Head Teaching Assistant

2020-2021 **Carnegie Mellon University**, Introduction to Computer Systems Teaching Assistant

2019-2020 **Carnegie Mellon University**, Principles of Imperative Computation Teaching Assistant

Service

2024-on **Women of the ACM Club Graduate Vice President**, Mentored several women undergraduates, helped organize events for the graduate student body and run logistics for the club.

2023–2023 **07-070, Teaching Assistant Panelist**, Was selected as a panelist for 07-070 (Teaching Techniques for Computer Science), a course teaching first-time TAs teaching techniques and DEI issues.

2022–2022 **Carnegie Mellon SCS Tour Guide**, Offered important information and advice about Carnegie Mellon's Computer Science program to prospective undergraduate students.

2020–2021 **Carnegie Mellon CS Education Group**, Participated in important discussions about computer science education with other passionate members of the CMU community.

2018–2021 **Carnegie Mellon Women@SCS**, Mentored several women underclassmen in the School of Computer Science and organized the Women in Systems panel, an event to highlight computer systems to women undergraduates.